

Data Center Impact Explorer

Final Presentation

DSCI 591, Summer 2026
Drexel University

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Image source: [Olivier Bonin/SLAC National Accelerator Laboratory/Wikimedia Commons](#)



Project Background

- The United States is home to the most data centers globally, with more on the way in the near future
 - Around 4,000 at present, and more than 1,500 in various stages of development¹
 - Used previously for cloud storage and digital streaming; recent boom with widespread AI adoption²
- Concerns regarding their impact on community wellbeing are beginning to emerge
 - Electricity and water usage, political implications, health concerns, etc.
- At present, there is no comprehensive, holistic representation of these concerns for public use

Project Goals

- Join multiple sources of data to investigate the relationship between data centers and local communities:

Are data centers located in areas already prone to **drought**, when their water demand might pose additional strain?

What are the conditions for **electricity production** and pricing where data centers are located?

What are existing **health concerns** in local communities that could be exacerbated by pollution from data centers?

What are the **political conditions** in areas with data centers, given local policymakers' influence on zoning and permitting?

- Present findings in clear visualizations hosted on public platform

Benefits and Stakeholders

- Information about effects of data center operation are dispersed or location-specific, so it is useful to bring together U.S.-wide info
 - Significant contribution is consolidation of different data sources
- Multiple possible stakeholders:
 - Community activists
 - Policymakers
 - Tech companies committed to responsible development
 - Health and infrastructure professionals

Data Sources & Cleaning

Sources:

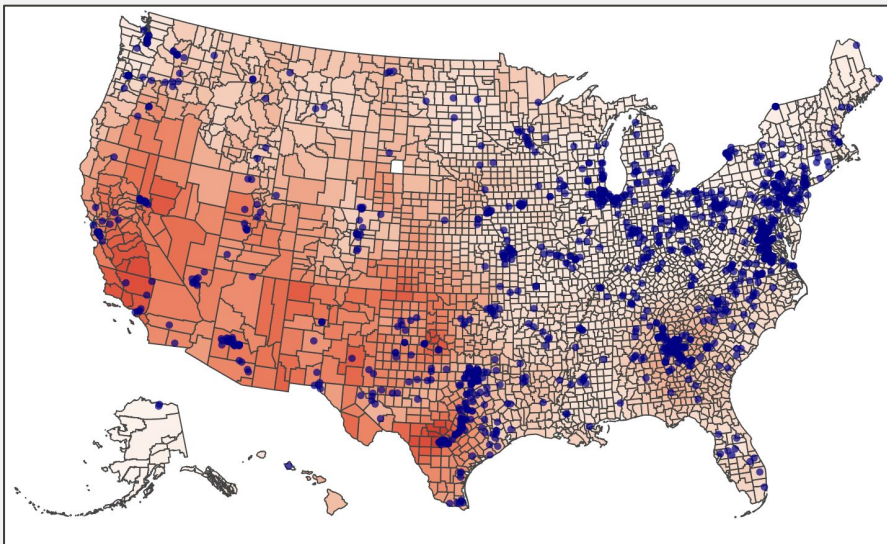
- FracTracker Alliance U.S. Data Center Database
- U.S. Energy Information Administration (EIA) Electricity Data
- America's Health Rankings (AHR)
- National Health Interview Survey (NHIS)
- U.S. Drought Monitor
- National Neighborhood Data Archive (NANDA)

Cleaning Steps:

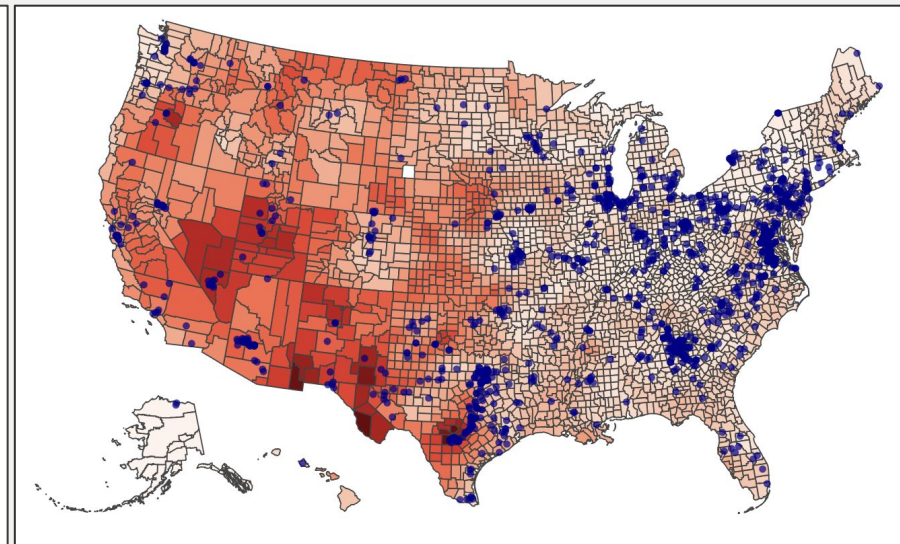
- Removing null values, string manipulation, checking for duplicates, and data-type conversion
- Paring down to useful and relevant columns
- Developed process to convert from ZIP to FIPS code to enable data combining, and aggregate number of data centers per FIPS code

Findings: Environment

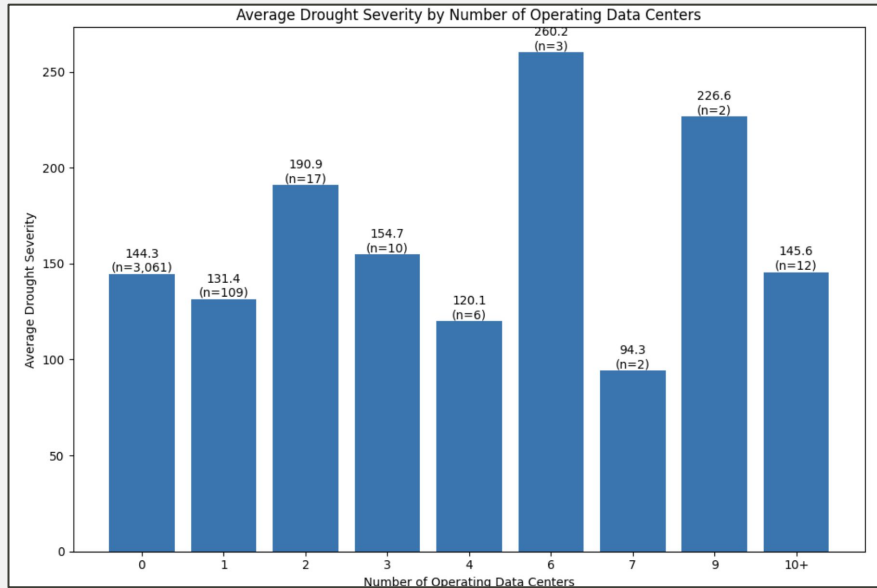
Average Drought Severity with Data Center Locations (2016 - 2021)



Average Drought Severity with Data Center Locations (2022 - 2026)



Findings: Environment

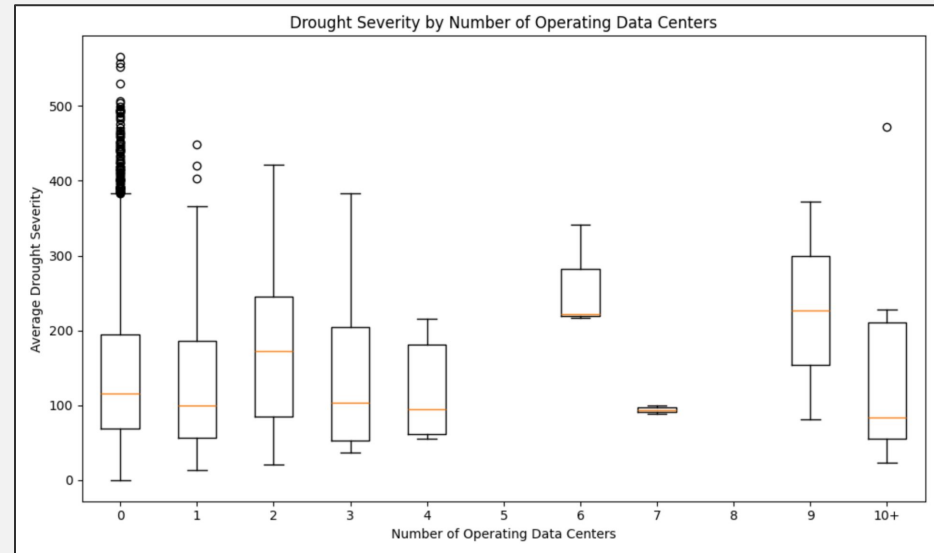


▲ **Fig. 1:** Not a clear correlation between the number of data centers in a county and the average drought of that county.

Note: no counties have 5 or 8 operating data centers.

▼ **Fig. 2:** Average varies. Boxes 0-4 look to be around the same range.

The categories with the most counties have the most similar box plots.



Findings: Energy

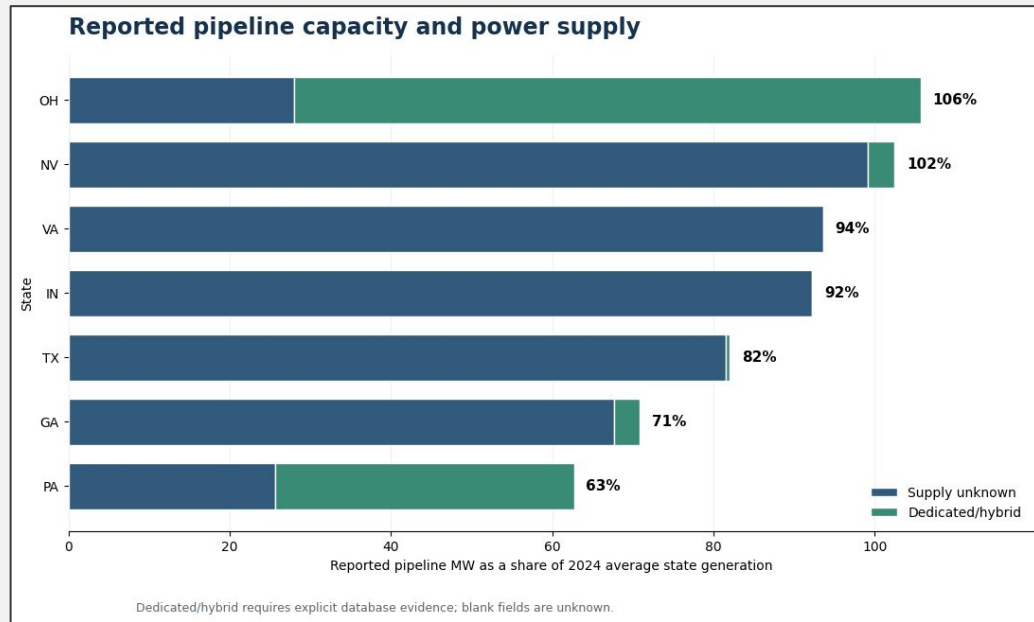


Fig. 1: Reported data-center pipeline capacity is approaching (or exceeding) average state electricity generation in several states.

62% of pipeline projects do not report MW, so the reported totals are incomplete.

$$\text{Pipeline share (\%)} = \frac{\text{Reported data-center pipeline MW}}{\text{State's 2024 annual generation MWh} \div 8,760} \times 100$$

Findings: Energy

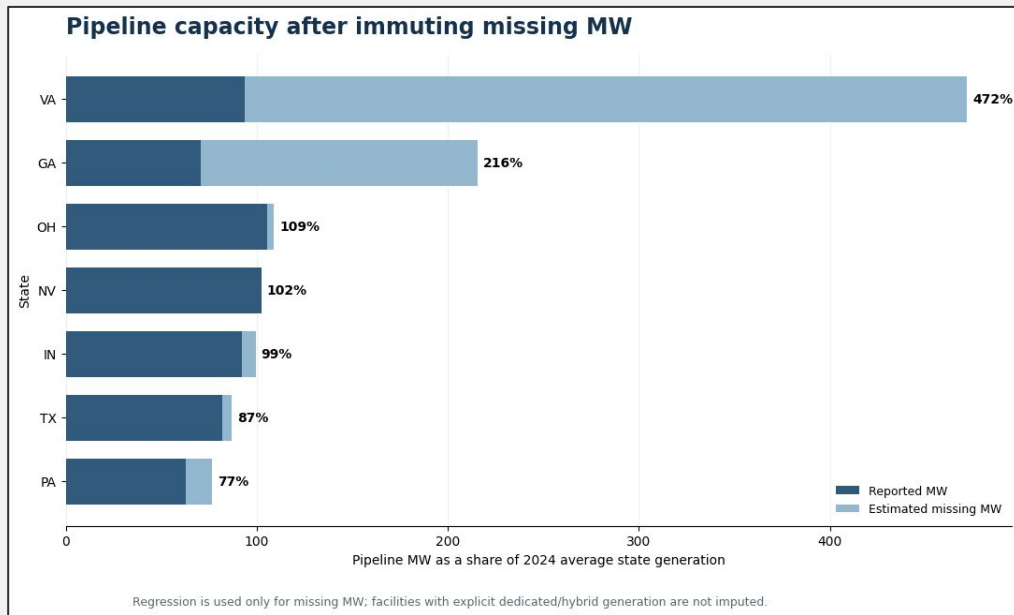


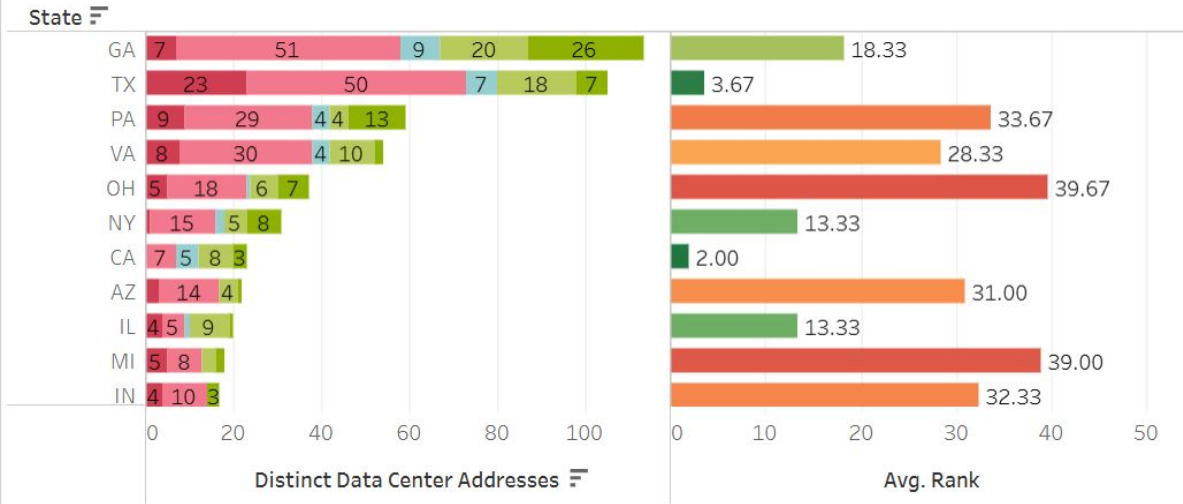
Fig. 1: Reported data-center pipeline capacity over average state electricity generation where Missing MW is imputed using linear regression on the size of the facility.

Data from FracTrackers U.S. Data Centers Tracker and U.S. Energy Information Administration (EIA) Electricity Data

Findings: Health

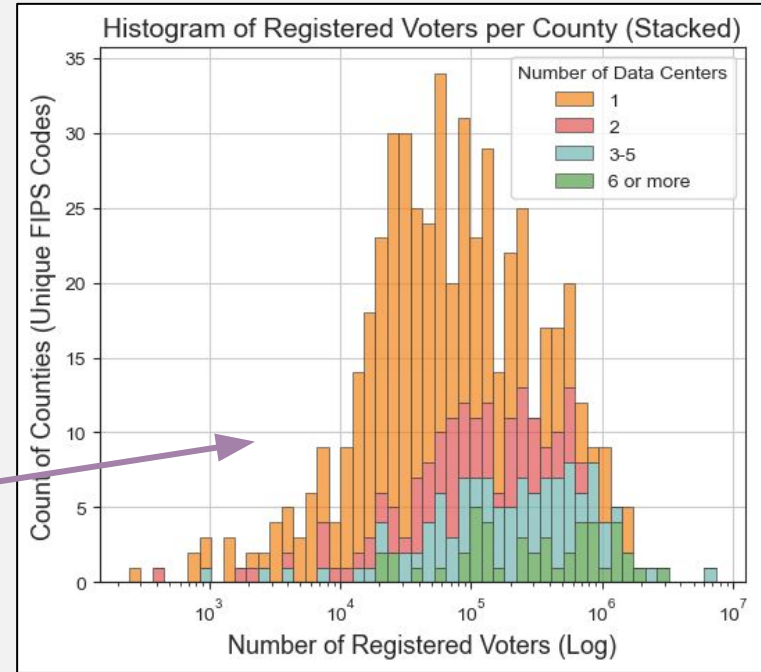
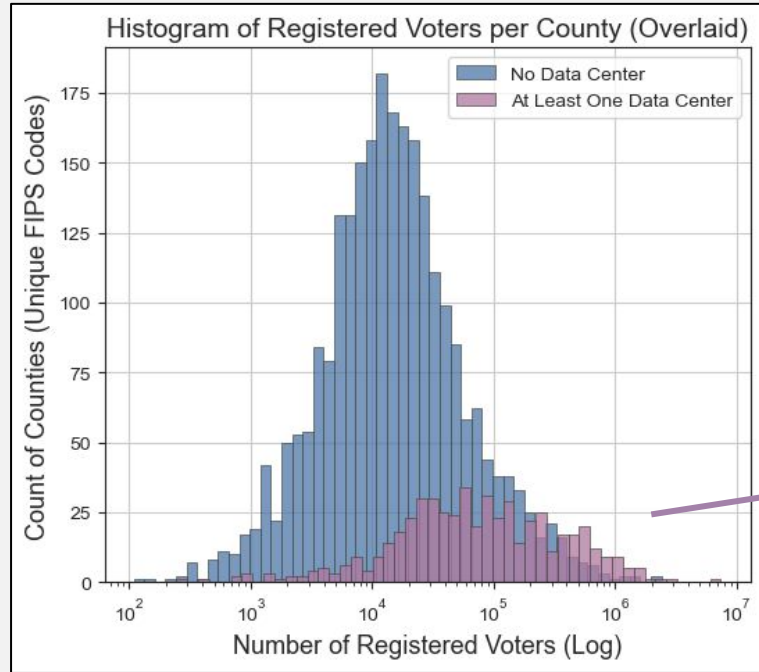
Number/Size of Data Centers and Average State Health Rank*

*across asthma, cancer, and cardiovascular disease rankings



- States with the **highest** number of data centers also have **middling or poor** health rankings
 - Notable exceptions for **Texas** and **California**
 - Varied amount from each of the **four U.S. Census Regions**
- Data centers in these states are also largely **hyperscale**

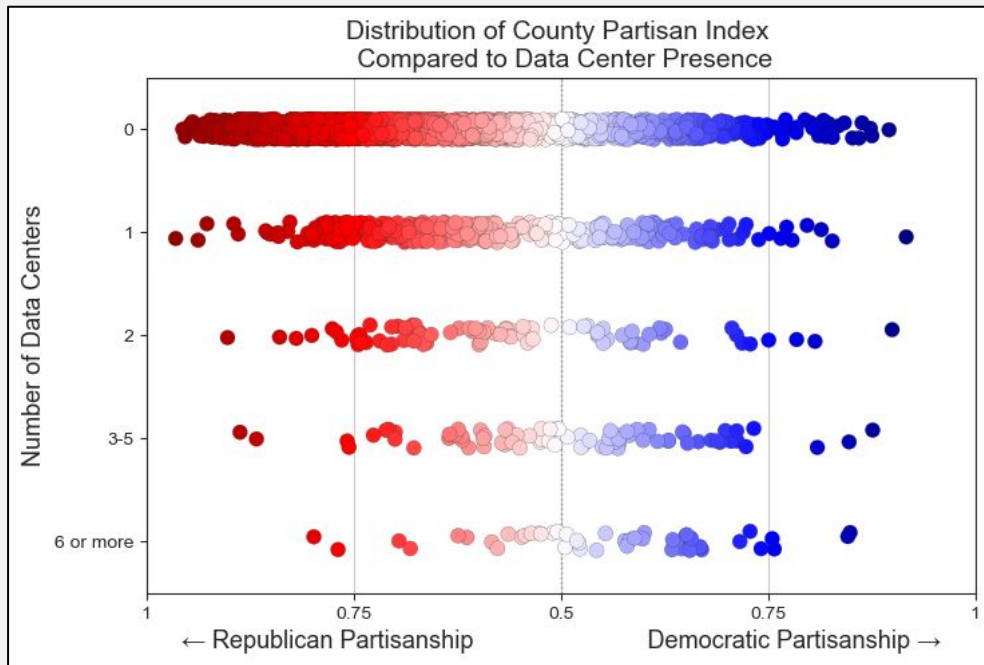
Findings: Politics



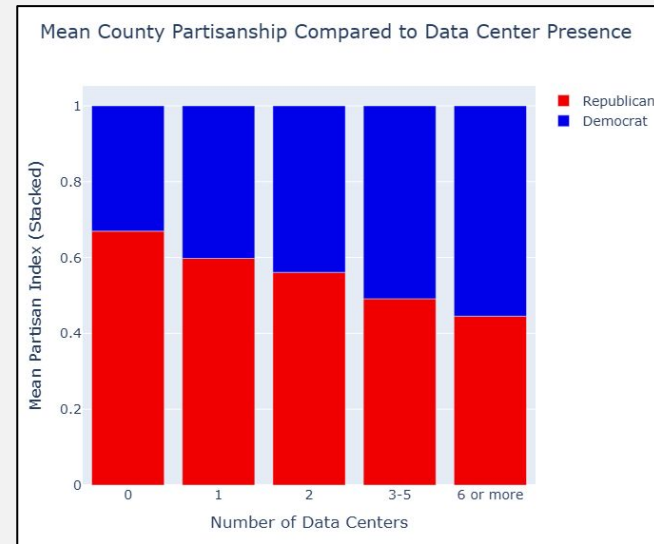
Including both operating and proposed data centers

Data from FracTrackers U.S. Data Centers Tracker and National Neighborhood Data Archive

Findings: Politics



summary



Number of Data Centers	Mean Republican Partisan Index	Mean Democrat Partisan Index
0	0.67	0.33
1	0.60	0.40
2	0.56	0.44
3-5	0.49	0.51
6 or more	0.45	0.55

Data from FracTrackers U.S. Data Centers Tracker
and National Neighborhood Data Archive

Limitations

- Data center dataset does not cover every site in the U.S., and is partially crowd-sourced; may be biased towards sites in higher-population areas
 - Other datasets with 2x+ more data centers exist but are paywalled or otherwise not easily accessible
 - Does not contain the year each datacenter became operational which limits time series analysis
- Supplementary datasets are not all up-to-date to current time, while many of the data centers in the dataset are just proposed
 - We were only able to examine existing conditions where data centers are/may be located rather than investigate direct impacts
- Our work thus far can serve as a broad summary/proof of concept

Future Work

- Finalize public page via Streamlit
 - Finishing touches on visualizations, information presentation, and data integration
 - Final GitHub push
- Maintain page as relevant data and studies get released
- Continue attempting partnerships with data center awareness groups
- Potentially publish process write-up in *Medium* or *Towards Data Science*



Streamlit

Medium

Data Center Impact Explorer Dashboard

Data centers are the backbone of cloud storage, digital streaming, and artificial intelligence, but concerns regarding the impact they may have on the surrounding communities are beginning to emerge. Here, we hope to specifically address the following four areas of interest and illustrate how the presence of a data center may impact the lives of the individuals currently residing there:

- **Drought:** data centers require vast amounts of water to cool their servers; are the communities they're located in already suffering drought conditions?
- **Electricity:** data centers also require electricity; how might this impact the cost of energy in the surrounding areas?
- **Health:** the pollutants released by data centers can exacerbate cardiovascular and lung conditions, as well as cancer; do members of the surrounding community tend to already have these ailments?
- **Politics:** data centers are divisive; could a data center being built in a community cause political friction in the community?

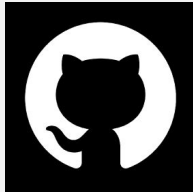
US Map, Data Center Locations, and Impacts

Please select a state to view how to see how a new data center might broadly impact residents!

Choose a U.S. state:

Select a state...

GitHub Repository Overview



github.com/agrace339/Data-Center-Impact-Explorer

Features:

- Detailed ReadMe, including instructions for acquiring data for replicability
- Scripts in Jupyter notebooks with text explanations
- Record of presentations and reports to tell complete story of research process

📁 Data	Initial commit	last month
📁 Presentations and Reports	Add folder with presentations and reports	5 hours ago
📁 Script	Final health data cleaning and investigation script	2 hours ago
📄 .gitignore	Update README.md	last month
📄 README.md	Update README.md - pres and reports	5 hours ago

README  

Data Center Impact Explorer

Drexel University MS in Data Science Capstone I

A data science capstone project analyzing the relationship between data centers and surrounding communities across the United States and investigating the potential impacts of data center development on local areas.

The project integrates multiple public datasets with data from the the FracTracker Alliance U.S. Data Centers Tracker to explore factors that may influence where data centers are built and existing conditions or issues within the surrounding areas that could potentially be exacerbated by data center development.

Conclusions

- Data centers have become a major issue over the past couple of years around the U.S., and will continue to be built.
- With this project, we wanted to see the impact of data centers on local communities
- **Findings**
 - ***Drought:*** No evidence to suggest data centers cause more severe drought, but should continue to be monitored for long-term impact.
 - ***Energy:*** Data centers are projected to consume more energy than was generated in 2024.
 - ***Health:*** Data centers are either currently operating or set to be built in areas with substantial relevant health concerns, which should be accounted for during the development process.
 - ***Politics:*** Data centers are often sited in high-population areas with more Democratic-leaning political views, with implications for public sentiment and policy responses.

References

- CDC. (2024). NCHS Urban-Rural Classification Scheme for Counties. *National Center for Health Statistics*. Retrieved from <https://www.cdc.gov/nchs/data-analysis-tools/urban-rural.html>
- Clary, W., Gomez-Lopez, I. N., Chenoweth, M., Gypin, L., Clarke, P., Noppert, G., Li, M., & Kollman, K. (2024). National Neighborhood Data Archive (NaNDA): Voter Registration, Turnout, and Partisanship by County, United States, 2004-2022 [Data set]. *Inter-university Consortium for Political and Social Research*.
<https://www.icpsr.umich.edu/web/ICPSR/studies/38506/versions/V2>
- Federal Communications Commission. (n.d.). *Federal Information Processing System (FIPS) Codes for States and Counties*. Retrieved July 28, 2026, from
<https://transition.fcc.gov/oet/info/maps/census/fips/fips.txt>
- FracTracker. (2026, July). U.S. Data Centers Tracker [Data set]. *FracTracker Alliance*.
<https://experience.arcgis.com/experience/5a4d072ad01449bba5698a80103fb909/page/Demographics>
- Han, Y., Wu, Z., Li, P., Wierman, A., & Ren, S. (2024). Health-Informed Computing: Estimating and Addressing the Public Health Impact of Data Centers. Retrieved from
<https://arxiv.org/abs/2412.06288>
- National Center for Health Statistics. (2024). 2024 NHIS Questionnaire, Datasets, and Documentation [Data set]. *U.S. Centers for Disease Control and Prevention*.
<https://www.cdc.gov/nchs/nhis/documentation/2024-nhis.html>
- National Center for Health Statistics. (2024). 2024 NHIS Questionnaire, Datasets, and Documentation [Codebook]. *U.S. Centers for Disease Control and Prevention*.
<https://www.cdc.gov/nchs/nhis/documentation/2024-nhis.html>
- Pate, C. A., Akinbami, L. J., Johnson, C., Hsu, J., & Zahran, H. S. (2025). Asthma and Allergy Comorbidity Among the US Population Aged 2 Years or Older, National Health Interview Survey, 2021. Public health reports (Washington, D.C. : 1974), 140(5-6), 540–550. <https://doi.org/10.1177/00333549251358658>
- Pavlinich, E. J. (2026). The Dangers of Data Centers. Environmental Health Project. Retrieved from <https://www.environmentalhealthproject.org/post/the-dangers-of-data-centers>
- Row Zero. (2025, March 6). *FIPS codes for all U.S. locations in a spreadsheet* [Data set]. Retrieved July 21, 2026, from <https://rowzero.com/datasets/fips-codes-lookup#zip-code-mappings>
- Stansbury, M., Marchese, K., Hardin, K., & Amon, C. (2025, June 24). Can US infrastructure keep up with the AI economy? Deloitte Insights. <https://www.deloitte.com/us/en/insights/industry/power-and-utilities/data-center-infrastructure-artificial-intelligence.html>
- Seets, S., & Radde, K. (2026). Most new data centers in the U.S. are coming to rural areas | Pew Research Center. In Pew Research Center.
<https://www.pewresearch.org/short-reads/2026/04/13/most-new-data-centers-in-the-us-are-coming-to-rural-areas/>
- United Health Foundation (2025). America's Health Rankings 2025 annual report [Data set]. *America's Health Rankings*.
<https://www.americashealthrankings.org/publications/reports/2025-annual-report>
- United Health Foundation. (2025). 2025 Annual Report: Measures table. *America's Health Rankings*.
https://assets.americashealthrankings.org/ahr_2025annual_measuresstable_final-web.pdf
- United States Census Bureau (n.d.). Geographic Levels. Guidance for Economic Census Geography Users. Retrieved from
<https://www.census.gov/programs-surveys/economic-census/guidance-geographies/levels.html>
- U.S. Drought Monitor. (n.d.). Data Tables [Data set]. *U.S. Drought Monitor*. Retrieved June 30, 2026, from <https://droughtmonitor.unl.edu/DmData/DataTables.aspx>
- U.S. Energy Information Administration (2025). *EIA Bulk Data* [Data set]. <https://www.eia.gov/opendata/>

Questions?

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